DARSINI LAKSHMIAH

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SUMMARY

Data Analyst with 4+ years delivering production analytics at Amazon, AARP, and NatWest driving ~9-10% engagement lift, ~20-35% query performance gains, and ~50% faster insights delivery across 87M+ transactions in grocery retail, digital marketing, and UK/EMEA banking.

WORK EXPERIENCE

AARP SERVICES INCORPORATED, Washington, DC, USA

Data Scientist Intern **September 2025 – March 2026**

- **Reduced manual campaign operations by ~25% by automating ~30 recurring marketing campaigns** previously executed in SAS using n8n workflows integrated with Databricks, PySpark, and Redshift APIs, enabling scheduled execution, automated reporting, and monitoring for marketing analytics teams.
- Analyzed **76M+** user interaction records from S3 and Redshift to understand engagement patterns across AARP digital platforms and developed a LightGBM-based recommendation model that improved campaign targeting across content categories, **increasing user engagement by ~10%.**
- Built an interactive marketing analytics dashboard using Databricks and Redshift to monitor key campaign KPIs (open rates, CTR, engagement trends), enabling marketing teams to quickly evaluate performance and optimize campaign strategy.
- **Improved accessibility of analytics insights by developing a LLaMA-powered analytics assistant** integrated with Databricks and Redshift, enabling marketing and product teams to explore campaign performance and user behavior through NL queries and reducing analytics turnaround time by **~50%**.

AMAZON, Austin, Texas, USA

Business Intelligence Engineer Intern **May 2025 - August 2025**

- **Enabled data-driven decisions for a $200M multi-banner grocery retail initiative** by developing a unified Customer Behavioral Analytics platform integrating online and in-store purchase data across Amazon Fresh and Whole Foods; analyzed lifetime value and purchasing patterns across **87M+** transactions and **530+** store locations
- Improved customer targeting and product experimentation capabilities by building a purchase-propensity prediction model using Python, SQL, and Amazon SageMaker; integrated predictions into analytics dashboards used by product and retail teams to evaluate feature launches and customer engagement trends.
- **Reduced analytics latency & improved reliability of operational dashboards by ~20%** by engineering scalable ETL pipelines with AWS Glue, Athena, S3, and Redshift, transforming raw transactional data into optimized SQL aggregations powering retail analytics.
- Provided actionable insights for new in-store digital features such as Dash Cart and Click-Checkout by building operational performance dashboards and analyzing store-level adoption patterns; presented findings to Product Managers, Data Scientists, and retail operations stakeholders guiding the Amazon Fresh, Whole Foods integrated store strategy.

NATWEST GROUP PRIVATE LIMITED, Chennai, India

Senior Data Analyst **Aug 2021 - June 2024**

- Built distributed ETL pipelines & scalable data models (Azure Databricks/PySpark, ADF) processing **1.5M+** daily banking transactions, improving query performance by **~35%** and powering product & risk reporting dashboards across banking channels.
- Strengthened AML and financial crime monitoring by developing NLP-driven anomaly detection model with TF-IDF feature engineering and XGBoost classification on transaction data ingested via Azure Data Factory into ADLS Gen2 and Snowflake identifying suspicious behavioral patterns across UK and EMEA operations in alignment with enterprise compliance standards.
- **Lifted UK campaign engagement by ~9% via A/B experimentation & segmentation** (Python statsmodels, Snowflake) on personalized credit card & savings offers; ensured data governance & regulatory compliance; delivered insights via Power BI dashboards to business stakeholders.

EDUCATION

- **George Washington University, Washington, DC** - Master of Science in Data Science, GPA - 4.0/4.0 **May 2026**
- **Madras Christian College, Chennai, India** - Bachelor of Science in Mathematics, CGPA - 8.50/10 **May 2021**

PROJECTS

AI Interview Coach | GitHub |Website

- Built a real-time multi-agent interview simulation using AutoGen, FastAPI, and GPT-4o three specialized AI agents (Interviewer, Evaluator, Candidate) conduct company-specific interviews over WebSocket with live feedback, webcam emotion/eye contact tracking via face-api.js, and verbatim Whisper transcription to generate performance reports with fluency scoring.

Dock Wise AI Maritime Port Intelligence Dashboard | GitHub

- Built a maritime supply chain intelligence platform ingesting live IMF Port Watch data across 117 US ports; delivers 14–28-day time series congestion forecasts using ARIMA, Prophet, and XGBoost, plus a RAG-based LangChain + Groq (LLaMA-3.3-70B) AI advisor with FAISS vector store for port and policy queries delivering real-time chokepoint and weather-risk recommendations.

NutriNav Food Access Intelligence App | GitHub | DevPost

- Top 3 of 35 teams - George Hacks × UN Reboot the Earth
- Built multilingual voice-first app connecting 75,000+ DC SNAP residents to real-time corner store inventory across DCCK's Healthy Corners network Groq Whisper auto-detects language, LLaMA 3.3 70B responds in kind; XGBoost grades products A-D via USDA Food Data.

TECHNICAL SKILLS

- **Programming Languages:** Python, R, SQL, SAS, PySpark
- **ML & AI:** Scikit-learn, TensorFlow, PyTorch, LangChain, LangGraph, AutoGen, LlamaIndex, Hugging Face
- **Cloud & Data Engineering:** AWS (Glue, Redshift, Athena, S3, Lambda, Bedrock), GCP BigQuery, Databricks, Snowflake, dbt, Airflow
- **Databases & Analytics:** MySQL, MongoDB, Apache Spark, Pandas, NumPy, A/B Testing, Causal Inference, KPI Measurement, Conversion Rate Analysis, Experimental Design.
- **Visualization & Tools:** Tableau, Power BI, QuickSight, Excel (Advanced), Git/GitHub, Docker, Flask.